

FrugalProver: A Self-Improving Math Agents System that Learns Its Boundaries

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Abstract

Today’s language models can already resolve some previously open mathematical problems, but at large, unpredictable cost and with errors that are hard to detect. We propose a compact, auditable multi-agent system that proves theorems under an explicit compute budget. A *budget oracle* estimates, for each problem, the effort (in tokens) a successful attempt warrants—avoiding both overspending on easy problems and premature termination on hard ones. A prover produces a candidate proof; skeptical verifiers audit it; a corrector repairs the gaps; only candidates passing every check reach a human reviewer. The system also improves over time by adopting new, sandbox-validated tactics. Its guiding premise—of interest beyond this system—is that the effort reasonable to spend on a proof, or on its verification, for a successful attempt can be estimated in advance from a small model’s internal activations. The pipeline runs entirely on open-weight models, making it reproducible and inexpensive.

Index Terms: AI for mathematics; multi-agent reasoning; efficient inference-time compute; automated theorem proving; geometry and topology of neural representations.

1 Introduction

Judging a problem’s difficulty before attempting it is routine for mathematicians and essential to allocating effort efficiently. We turn this intuition into a single control variable—the per-problem token budget—and ask whether a multi-agent system can predict it, allocate it well, and improve at both. The system builds on established propose-check-repair pipelines [4, 5, 8–12] and tool use [6], keeping the solver offline until an independent verifier validates each attempt.

Architecture. Every call to the backbone first passes the *budget oracle* $\mathcal{O}$, which predicts a budget $\hat{B}(x)$ and caps the response length. A prover produces a candidate proof; an ensemble of $k \geq 3$ skeptical verifiers audits it; an offline corrector repairs the gaps; the loop iterates until the verifiers concur. Accepted candidates pass to a *human gate*—the single point of human sign-off. Independently, an *orchestrator* proposes and sandbox-tests improved tactics before deployment.

Formulation. Let $p_i(B)$ be the probability of solving x_i within B tokens. Given a fixed total budget, the oracle allocates each additional token where it yields the greatest marginal gain in success [7]: $\max_{\{B_i\}} \sum_i p_i(B_i)$ subject to $\sum_i B_i \leq B_{\text{tot}}$. It predicts effort from two inexpensive signals—surface features of the statement $\phi(x)$ and a small model’s internal activations $g_\ell(x)$ —via $\hat{B}(x) = f_\theta(\phi(x), g_\ell(x))$, updated online as outcomes accrue.

1.1 Hypotheses

- H1** *Effort is predictable.* The internal signal g_ℓ , with simple surface features, predicts the required budget substantially better than a length baseline, and carries measurable information about effort, $\hat{I}(g_\ell; B^*) > 0$.
- H2** *Allocation improves efficiency.* Budget-aware allocation solves more problems per unit of compute than uniform or length-based budgets, raising the entire success-versus-compute curve.
- H3** *Self-improvement is safe.* Across rounds, online tuning and human-approved tactic updates increase problems-solved-per-token, while the human gate prevents any rise in the rate of wrongly accepted proofs.

1.2 Goals

- G1** *System.* Implement the budget-gated pipeline—prover, $k \geq 3$ verifiers, corrector, human gate, and the oracle through which every call passes—with complete logging; the system runs on open-weight models [13].
- G2** *Analysis.* Train f_θ and examine *why* it works, characterizing the geometry and topology of the model’s internal difficulty representation—its intrinsic dimension [2] and informativeness about effort [1].
- G3** *Evaluation and governance.* Report efficiency, safety, and self-improvement; release a reproducible toolkit and a human-audited case study on well-known but tractable problems.

1.3 Validation Protocol

Datasets. Competition mathematics with known answers—the MATH benchmark [3] and harder olympiad problems—allowing automatic grading of final answers, with a human auditor checking reasoning on a sample and on a few open-ended problems. The predictor is lightweight (a probe plus an allocation rule) and needs no backbone retraining.

Metrics. Problems-solved-per-token (area under the success-versus-compute curve); tokens to first valid solution; budget-prediction accuracy; wrongly accepted proofs at the gate; improvement across rounds; and geometric/topological measures of the difficulty representation.

Baselines. Uniform budget; length-based budget; repeated sampling / self-consistency [8] at matched compute; reduced verifier count; and the system with the oracle disabled.

Timeline. Online weeks build the harness and a fixed-budget baseline; week 1 trains the predictor and runs the allocation experiments; week 2 adds self-improvement and the geometry/topology analysis, with first results on the last day.

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